

Basic Linux Commands

Usefullink- <https://itworkshopktu2024.blogspot.com/2024/11/familiarization-of-basic-linux-commands.html>

1. Do the following in the order given
 - a) Create a directory EV4. (***mkdir ev4***)
 - b) Navigate to that directory (***cd ev4***)
 - c) Create a directory with your roll number
 - d) Navigate to that
 - e) Type the following commands and write the resultant directory path(use ***pwd*** if required) . Also pen down your understanding of the result
 - i. ***cd johan@johan MINGW64 ~***
go to directory
 - ii. ***cd - /c/Users/johan/EV4/32***
go to previous directory
 - iii. ***cd . johan@johan MINGW64 ~ /EV4/32***
stay in current directory
 - iv. ***cd .. johan@johan MINGW64 ~/EV4***
move to parent directory
 - v. ***cd ~ johan@johan MINGW64 ~***
go to home directory of current user
 - vi. ***cd / johan@johan MINGW64 /***
go to root directory of system
 - vii. ***ls -l johan@johan MINGW64 ~***
to list files with detailed information
 - viii. ***cd media bash: cd: media: No such file or directory***
move into the folder named media
 - ix. ***cd johan@johan MINGW64 ~***
takes to home directory
 - x. ***pwd /c/Users/johan***
present working directory
 - xi. ***cd media bash: cd: media: No such file or directory***
 - xii. ***cd /media bash: cd: /media: No such file or directory***
moves to the media folder located inside root directory
 - xiii. ***ls -l johan@johan MINGW64 ~***
to list fails with detailed information
 - xiv. ***ls -al johan@johan MINGW64 ~***
showing all files including hidden ones
 - xv. ***cd ~/ev4/<ur roll number> johan@johan MINGW64 ~/EV4/32***
go to the folder roll no 32 which is inside EV4
 - xvi. ***mkdir emptydummy johan@johan MINGW64 ~/dummy***
create new dictionary named emptydummy
 - xvii. ***mkdir dummy1 johan@johan MINGW64 ~/dummy***
create new dictionary dummy1

- xviii. *cd dummy johan@johan MINGW64 ~/dummy/dummy*
changes working directory to folder named dummy
- xix. *touch file1 johan@johan MINGW64 ~/dummy/dummy*
create a new file named file1 inside the current working directory
- xx. *touch file2 johan@johan MINGW64 ~/dummy/dummy*
Created a new empty file named 'file1' inside the current working directory
- xxi. *ls -l -rw-r--r-- 1 johan 197609 0 Feb 9 00:08 file1*
-rw-r--r-- 1 johan 197609 0 Feb 9 00:08 file2
- xxii. *rm -i file2 rm -i file2 rm: remove regular empty file 'file2'? y*
Deletes the file named "file2" after asking for confirmation.
- xxiii. *ls -l johan@johan -rw-r--r-- 1 johan 197609 0 Feb 9 00:08 file1*
Displayed all the files
- xxiv. *cd .. johan@johan MINGW64 ~/dummy*
Moves to parent directory('rollno_29')
- xxv. *rm emptydummy rm: cannot remove 'emptydummy': Is a directory*
Attempts to remove directory "emptydummy", but results in error since it is used for files.
- xxvi. *rmdir emptydummy johan@johan MINGW64 ~/dummy*
only empty dirs removed with rmdir
- xxvii. *rmdir dummy rmdir: failed to remove 'dummy': Directory not empty*
will give an error since not empty
- xxviii. *rm -r dummy johan@johan MINGW64 ~/dummy*
Delete the directory 'dummy' along with all the files inside it

2. **cat >file1.txt --** You can use cat to create a file and input text directly from the terminal. Type the content '**My first line**', and press CTRL+D to save and exit
3. **cat >file2.txt --** Type the content '**Hello Second line**', and press CTRL+D to save and exit
- 4.
5. **cat > file3.txt --** Write '**Hello line**' as input and save the file
6. **cat file1.txt file2.txt > file_combined.txt --** > overwrites, >> appends
7. **cat file_combined.txt --** Need not type the entire filename...Write file_c and press Tab to see how it autocompletes
8. **cat file3.txt >> file_combined.txt**
9. **cat file_combined.txt**
10. **grep -i hello file***
11. **cp file1.txt ~/ev4**
12. **mv file_combined.txt combined --** check new file using **ls -l**

Change permissions → chmod

You can do this in two ways.

Method A: Symbolic mode (easy to read)

Examples

1. Give execute permission to owner: ex: **chmod u+x file.sh**
2. Remove write permission from group: ex: **chmod g-w file.txt**
3. Add read permission to everyone: ex: **chmod a+r file.txt**
4. Set exact permissions: ex: **chmod u=rwx,g=rx,o=r myfile**

Method B: Numeric (octal) mode (most used)

Permission values for rwx = 421

Examples

1. Owner: rwx, Group: r-x, Others: r-- => `chmod 754 file.txt`
2. Read/write for owner only: => `chmod 600 file.txt`

Permissions meaning differ with ref to files and directories-

	Permission	File	Directory
	r	read file	list files (ls)
	w	modify file	create/delete files
13. <i>chmod u+x combined</i>	x	run file	enter directory (cd)
--Grant execute permission to owner. Check the new permission using <i>ls -l combined</i>			
14. <i>chmod g-r combined</i>	-- Remove read permission from group		
15. <i>chmod 777 combined</i>	-- giving rwx= 111=7, full permission to all user, group and others		
16. <i>sudo useradd alice</i>	-- new user created using sudo super user		
17. <i>sudo passwd alice</i>	-- set new password using passwd		
18. <i>sudo userdel alice</i>			

If in a network server, write command can work like a "chat" with someone logged into the same system(server)

The write command sends a real-time message to another user.

Both the sender and receiver must be logged into the same system.

The message is displayed directly on the receiver's terminal

Syntax : `write username [tty]`

username: The name of the user you want to send the message to.

tty (optional): Specifies the exact terminal session of the user (useful if the user has multiple sessions open).

Ex: ***write alice***

There is also an option for the user to enable/block messaging using ***mesg y*** or ***mesg n***